



Cyberster Blue Team Internship

Week 12 Report

Insider Threat Investigation: Confidential Data Leak at M57.biz

Prepared by: Muhammad Ubaid

Track: Blue Team Operations

Phase: Digital Forensics and Threat Intelligence

Submitted To: Sir Abdullah Zia (SOC Training Instructor)

Date: 23 May 2026

Core Competencies:

**Volatile Memory Forensics & Kernel Triage | Process Tree
Hierarchy Mapping | Live Network Socket Extraction | Registry
UserAssist Execution Tracking | Command-Line Console Buffer
Recovery | Insider Threat & Data-Exfiltration Mapping |
Multi-Layer Evidential Synthesis & Correlation**

1. Overview

This week is the Phase Two capstone. The job was to investigate Jo, an employee of [M57.biz](#) (a small patent search firm), who is suspected of intellectual property theft. Three evidence sources had to be analyzed together:

- Disk image: jo-2009-11-24.E01
- RAM dump: jo-2009-11-16.mddramimage (765.99 MB)
- Two USB images: jo-favorites-usb-2009-12-11.E01 and jo-work-usb-2009-12-11.E01

Tools used: Autopsy 4.22.1, Volatility 3 (2.28.0), Volatility 2.6, Registry Explorer 2026.5.0, LECmd 2026.5.0, CertUtil.

The investigation needed to answer three questions:

- What is Jo's job designation?
- What crime was committed?
- How was the data exfiltrated?

Short answers: Patent Researcher | IP Theft / Corporate Espionage | Two-channel exfiltration via USB + SMB share.

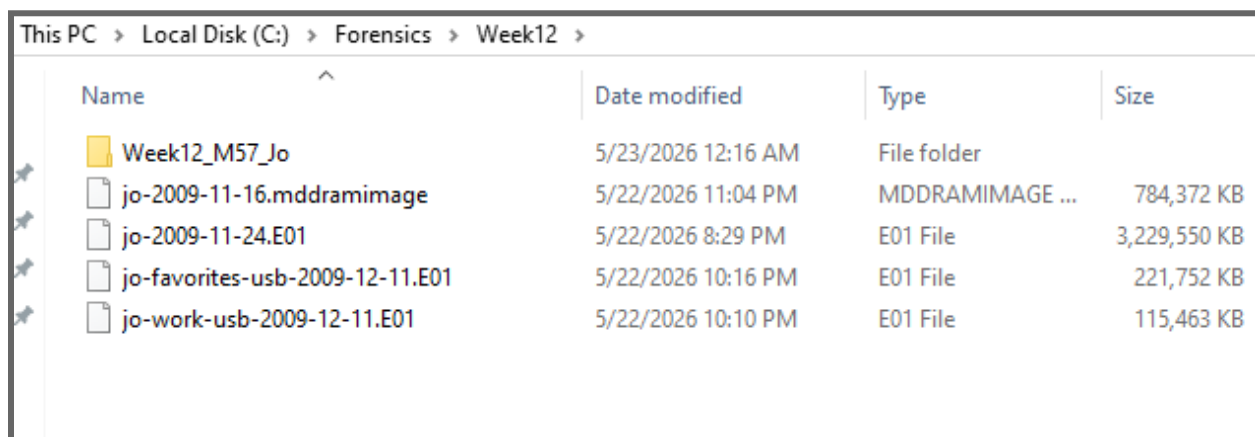
2. Investigation Anchors

#	Parameter	Value
1	Organisation	M57.biz (4 employees: CEO Pat McGoo, IT Admin, researchers Jo & Charlie)
2	Subject	Jo
3	OS	Windows XP Pro SP3, build 2600, x86
4	Hostname	M57-JO
5	Timezone	Pacific Standard Time (UTC-8, Bias=480)
6	Local IP	192.168.1.102
7	Other LAN host	192.168.1.104 (SMB session)
8	RAM capture time	2009-11-17 00:22:31 UTC
9	Disk shutdown	2009-11-25 05:46:24 UTC
10	USB acquired	2009-12-11

3. Part A — Triage and Subject Profiling (Days 81–82)

3.1 Working Folder Setup

Working folder: C:\Forensics\Week12\ containing all evidence images, extracted hives, LNK files, Volatility output, and USB extracts.



This PC > Local Disk (C:) > Forensics > Week12 >

Name	Date modified	Type	Size
Week12_M57_Jo	5/23/2026 12:16 AM	File folder	
jo-2009-11-16.mddramimage	5/22/2026 11:04 PM	MDDRAMIMAGE ...	784,372 KB
jo-2009-11-24.E01	5/22/2026 8:29 PM	E01 File	3,229,550 KB
jo-favorites-usb-2009-12-11.E01	5/22/2026 10:16 PM	E01 File	221,752 KB
jo-work-usb-2009-12-11.E01	5/22/2026 10:10 PM	E01 File	115,463 KB

3.2 Pre-Ingest Hash Verification (CertUtil)

File	MD5
jo-2009-11-24.E01	4474b818d2ad84beb74fbd7077d59e97
jo-2009-11-16.mddramimage	50c49185e254751caa0a9eadfa8fc635
jo-favorites-usb-2009-12-11.E01	8cdc12e30af14e19533c58b3ffe840b5
jo-work-usb-2009-12-11.E01	f9408bfcd292a7d8d60928a42806046f

SHA-256 of primary disk:

8fcfeb7a3e16e065e7e9112887b3b5c2047dcdf8706eeae188c5bcd3ce63ccd5

```

C:\Users\Ubaid>certutil -hashfile C:\Forensics\Week12\jo-2009-11-24.E01 MD5
MD5 hash of C:\Forensics\Week12\jo-2009-11-24.E01:
4474b818d2ad84beb74fbd7077d59e97
CertUtil: -hashfile command completed successfully.

C:\Users\Ubaid>certutil -hashfile C:\Forensics\Week12\jo-work-usb-2009-12-11.E01 MD5
MD5 hash of C:\Forensics\Week12\jo-work-usb-2009-12-11.E01:
f9408bfcd292a7d8d60928a42806046f
CertUtil: -hashfile command completed successfully.

C:\Users\Ubaid>certutil -hashfile C:\Forensics\Week12\jo-favorites-usb-2009-12-11.E01 MD5
MD5 hash of C:\Forensics\Week12\jo-favorites-usb-2009-12-11.E01:
8cdc12e30af14e19533c58b3ffe840b5
CertUtil: -hashfile command completed successfully.

C:\Users\Ubaid>certutil -hashfile C:\Forensics\Week12\jo-2009-11-16.mddramimage MD5
MD5 hash of C:\Forensics\Week12\jo-2009-11-16.mddramimage:
50c49185e254751caa0a9eadfa8fc635
CertUtil: -hashfile command completed successfully.

```

3.3 Autopsy Case Creation

- Case Name: M57-Jo
- Base Directory: C:\Forensics\Week12\
- Case Number: 001
- Examiner: Muhammad Ubaid
- Organisation: Cyberster Blue Team Internship

3.4 Keyword List

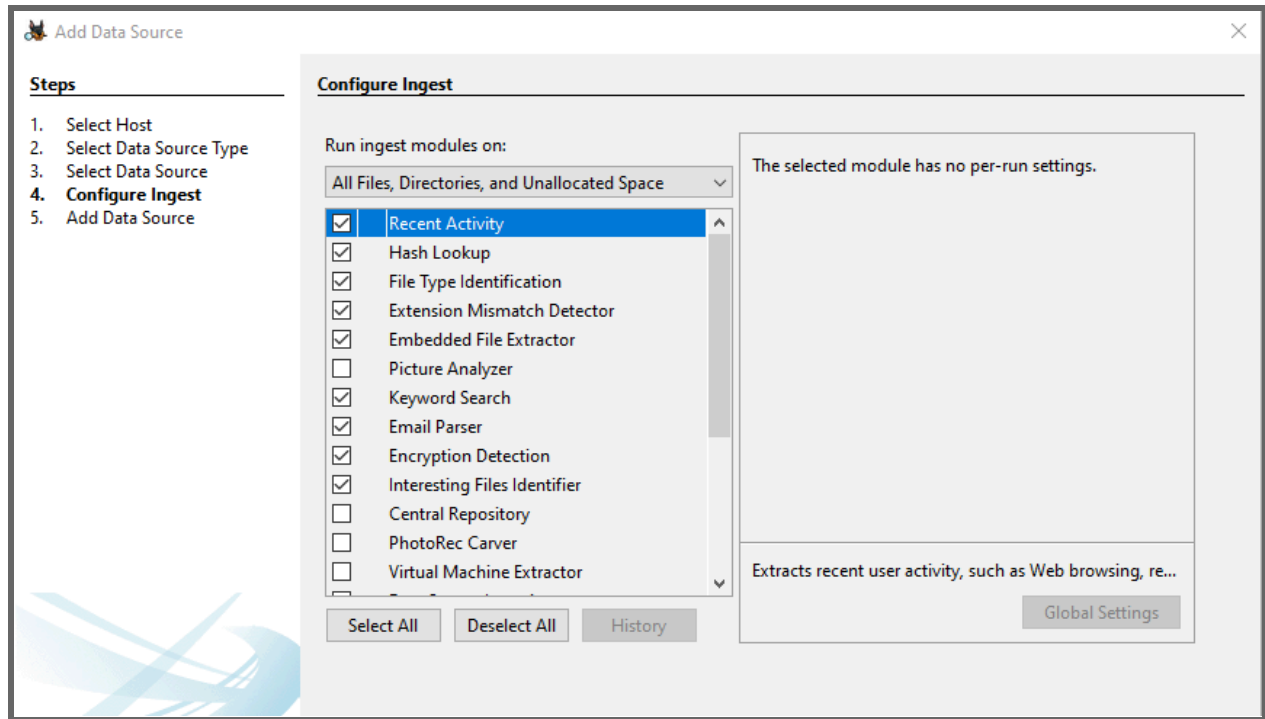
Created keyword list "Jo" with 4 exact-match terms: m57, patent, jo, confidential.

Keywords:	
Keyword	Keyword Type
m57	Exact Match
Patent	Exact Match
Jo	Exact Match
Confidential	Exact Match

Global Keyword Search Settings showing Jo list

3.5 Ingest Modules Enabled

Recent Activity, Hash Lookup, File Type ID, Extension Mismatch Detector, Embedded File Extractor, Picture Analyzer, Keyword Search, Email Parser, Encryption Detection, Interesting Files Identifier. PhotoRec Carver re-run later on the work USB only.

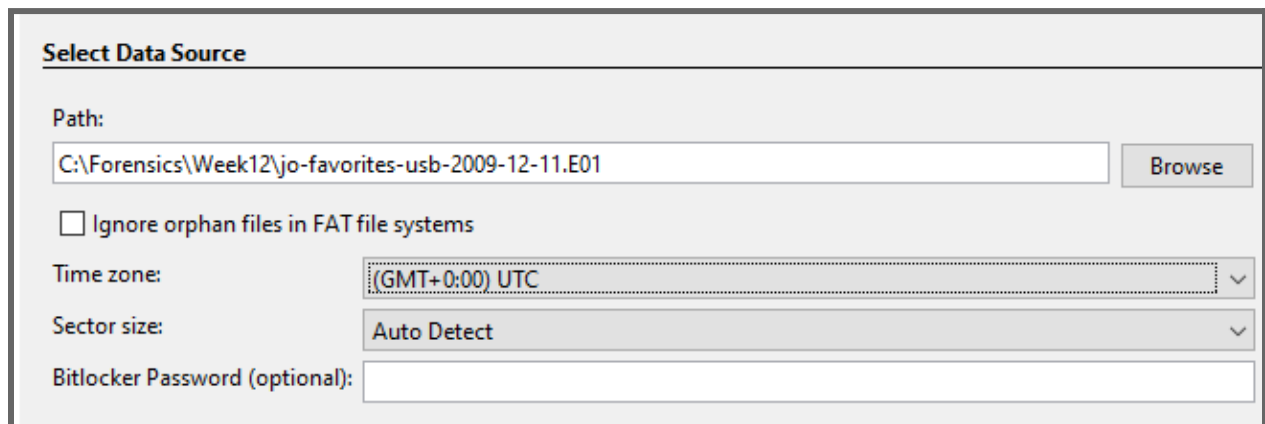


Configure Ingest Modules window

3.6 Data Sources Added

All three E01 images added with timezone set to (GMT+0:00) UTC . After ingest:

- Keyword Hits: 115,083
- Web History: 3,373
- USB Device Attached: 9
- Installed Programs: 96
- Encryption Detected: 7



Add Data Source: favourites USB

Select Data Source

Path:

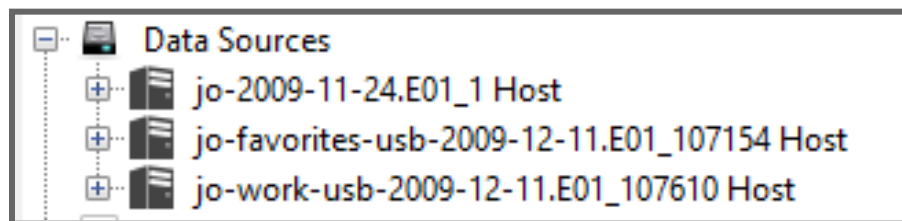
☐ Ignore orphan files in FAT file systems

Time zone:

Sector size:

Bitlocker Password (optional):

Add Data Source: work USB



Data Sources panel after all three ingests complete

3.7 Data Source Integrity Verification

Source	Calculated MD5	Stored MD5	Result
Primary disk	274ada9b6b120f9b7a2192c27e115244	same	✓ Verified
Favourites USB	ad1d03cbdb8d81e918899479360f50dc	same	✓ Verified
Work USB	8f23279deb398c3245829a98bc8fc1bd	same	✓ Verified

Data Source Verification Results for jo-favorites-usb-2009-12-11.E01

- Result: verified
- MD5 hash verified
 - Calculated hash: ad1d03cbdb8d81e918899479360f50dc
 - Stored hash: ad1d03cbdb8d81e918899479360f50dc

Data Source Verification: favourites USB

Data Source Verification Results for jo-work-usb-2009-12-11.E01

- Result: verified
- MD5 hash verified
 - Calculated hash: 8f23279deb398c3245829a98bc8fc1bd
 - Stored hash: 8f23279deb398c3245829a98bc8fc1bd

Data Source Verification: work USB

3.8 SOFTWARE Hive Baseline

Extracted from /WINDOWS/system32/config/software. Loaded in Registry Explorer → Microsoft\Windows NT\CurrentVersion:

Value	Data
ProductName	Microsoft Windows XP
CurrentBuildNumber	2600
BuildLab	2600.xpsp_sp3_gdr.090804-1435
CSDVersion	Service Pack 3
RegisteredOrganization	<u>M57.biz</u>
RegisteredOwner	JO
InstallDate	2009-11-20 14:34:50 UTC

Values		
Name	Type	Value
ProductName	REG_SZ	Microsoft Windows XP
RegDone	REG_SZ	(value not set)
RegisteredOrganization	REG_SZ	M57.biz
RegisteredOwner	REG_SZ	JO
SoftwareType	REG_SZ	SYSTEM
CurrentVersion	REG_SZ	5.1
CurrentBuildNumber	REG_SZ	2600
BuildLab	REG_SZ	2600.xpsp_sp3_gdr.090804-1435

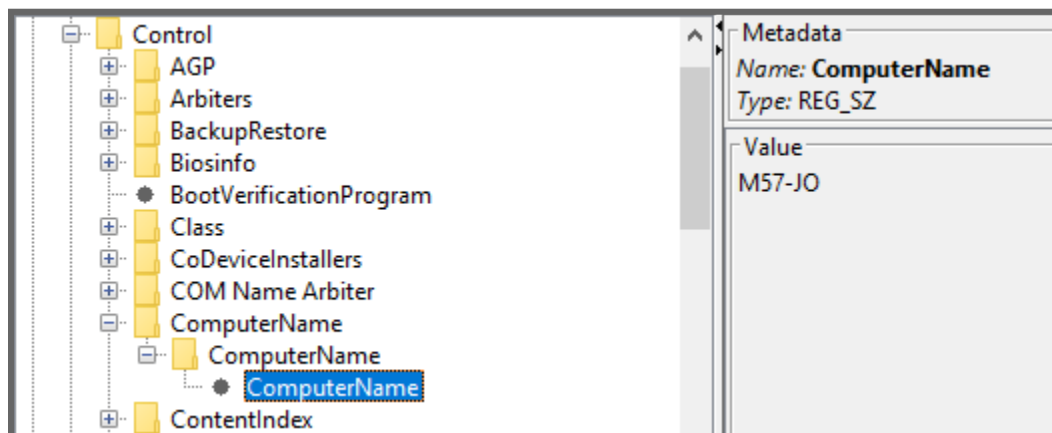
Registry Explorer: SOFTWARE hive CurrentVersion key

3.9 SYSTEM Hive Baseline

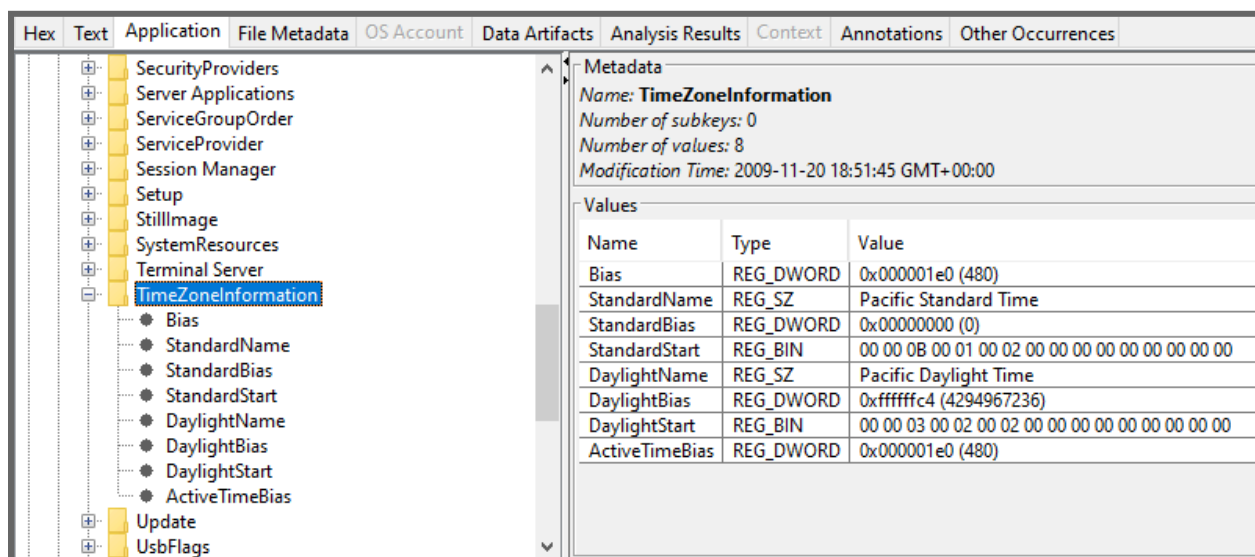
ControlSet001\Control\ComputerName\ComputerName → M57-JO

ControlSet001\Control\TimeZoneInformation:

- Bias: 480 (UTC-8)
- StandardName: Pacific Standard Time
- ActiveTimeBias: 480



Registry Explorer: SYSTEM hive ComputerName



Registry Explorer: TimeZoneInformation key

3.10 OS Account — Jo

Field	Value
Login	jo
SID	S-1-5-21-606747145-1547161642-1644491937-1003
Created	2009-11-20 19:35:57 UTC
Last Login	2009-11-24 01:28:50 UTC
Login Count	6
Administrator	True
Home	/Documents and Settings/Jo

Jo is a local administrator — this is what allowed her to run mdd_1.3.exe later.

Name	S	C	O	Login Name	Host	Scope	Realm Name	Creation Time
S-1-5-21-606747145-1547161642-1644491937-500			0	administrator	jo-2009-1...	Local		
S-1-5-21-606747145-1547161642-1644491937-1003			0	jo	jo-2009-1...	Local		
S-1-5-18				SYSTEM	jo-2009-1...	Local	NT AUTHORITY	
S-1-5-19				LOCAL SERVICE	jo-2009-1...	Local	NT AUTHORITY	
S-1-5-80-324959683-3395802011-921526492-919036			0		jo-2009-1...	Local	NT SERVICE	
S-1-5-20				NETWORK SERVICE	jo-2009-1...	Local	NT AUTHORITY	

Hex	Text	Application	File Metadata	OS Account	Data Artifacts	Analysis Results	Context	Annotations	Other Occurrences
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Basic Properties

Login: jo
Full Name:
Address: S-1-5-21-606747145-1547161642-1644491937-1003
Type:
Creation Date:
Object ID: 3014

jo-2009-11-24.E01_1 Host Details

Home Directory: /Documents and Settings/Jo

Realm Properties

Name: Unknown
Address: S-1-5-21-606747145-1547161642-1644491937
Scope: Local
Confidence: Inferred

Autopsy OS Accounts panel with Jo selected

3.11 Installed Software (Key Items)

Program	Significance
Mozilla Firefox 3.5.5	Primary browser, source of 3,373 history entries
MozRepl extension	Lets scripts drive Firefox via TCP — used by patentauto.py
Python 2.6.4 (downloaded 2009-11-23)	Runtime for patentauto.py
Java SE 6	Required by diit-1.5.jar
QuickTime (downloaded 2009-11-24)	Plays .m4v/.mov on work USB
AVG 9.0	Active AV — 11 AVG processes in pslist
OpenOffice 3	Office suite (no MS Office installed)
ManTech mdd_1.3.exe	Memory acquisition tool — run from USB drive E:

3.12 Jo's Profile — My Documents / Downloads / Desktop

Downloads folder (3 files + Zone.Identifier ADS each):

File	Created (UTC)	Size
QuickTimeInstaller.exe	2009-11-24 22:08:37	32.5 MB
python-2.6.4.msi	2009-11-23 18:20:14	14.9 MB
STU.doc	2009-11-24 22:01:53	28.5 KB

△ Name	S	C	O	Modified Time	Change Time	Access Time
Downloads				2009-11-25 03:08:50 PKT	2009-11-25 03:08:50 PKT	2009-11-25 03:08:50 PKT
My Music				2009-11-21 01:00:19 PKT	2009-11-21 01:00:19 PKT	2009-11-24 06:28:19 PKT
My Pictures				2009-11-21 01:00:19 PKT	2009-11-21 01:00:19 PKT	2009-11-24 06:28:19 PKT
[current folder]				2009-11-23 23:20:14 PKT	2009-11-23 23:20:14 PKT	2009-11-25 05:25:16 PKT
[parent folder]				2009-11-24 02:55:48 PKT	2009-11-24 02:55:48 PKT	2009-11-25 05:23:57 PKT
desktop.ini	▽		0	2009-11-21 01:00:19 PKT	2009-11-21 01:00:19 PKT	2009-11-25 05:25:16 PKT

My Documents listing

△ Name	S	C	O	Modified Time	Change Time	Access Time	Created Time
QuickTimeInstaller.exe	▽		0	2009-11-25 03:08:50 PKT	2009-11-25 03:09:05 PKT	2009-11-25 03:09:05 PKT	2009-11-25 03:08:37 PKT
QuickTimeInstaller.exe:Zone.Identifier			0	2009-11-25 03:08:50 PKT	2009-11-25 03:09:05 PKT	2009-11-25 03:09:05 PKT	2009-11-25 03:08:37 PKT
STU.doc	▽		0	2009-11-25 03:01:53 PKT	2009-11-25 03:01:53 PKT	2009-11-25 03:01:53 PKT	2009-11-25 03:01:51 PKT
STU.doc:Zone.Identifier			0	2009-11-25 03:01:53 PKT	2009-11-25 03:01:53 PKT	2009-11-25 03:01:53 PKT	2009-11-25 03:01:51 PKT
[current folder]				2009-11-25 03:08:50 PKT	2009-11-25 03:08:50 PKT	2009-11-25 03:08:50 PKT	2009-11-23 23:20:14 PKT
[parent folder]				2009-11-23 23:20:14 PKT	2009-11-23 23:20:14 PKT	2009-11-25 05:25:16 PKT	2009-11-21 00:58:51 PKT
python-2.6.4.msi	▽		0	2009-11-23 23:23:03 PKT	2009-11-23 23:23:03 PKT	2009-11-24 06:29:33 PKT	2009-11-23 23:20:14 PKT
python-2.6.4.msi:Zone.Identifier			0	2009-11-23 23:23:03 PKT	2009-11-23 23:23:03 PKT	2009-11-24 06:29:33 PKT	2009-11-23 23:20:14 PKT

Downloads folder showing 3 files + ADS

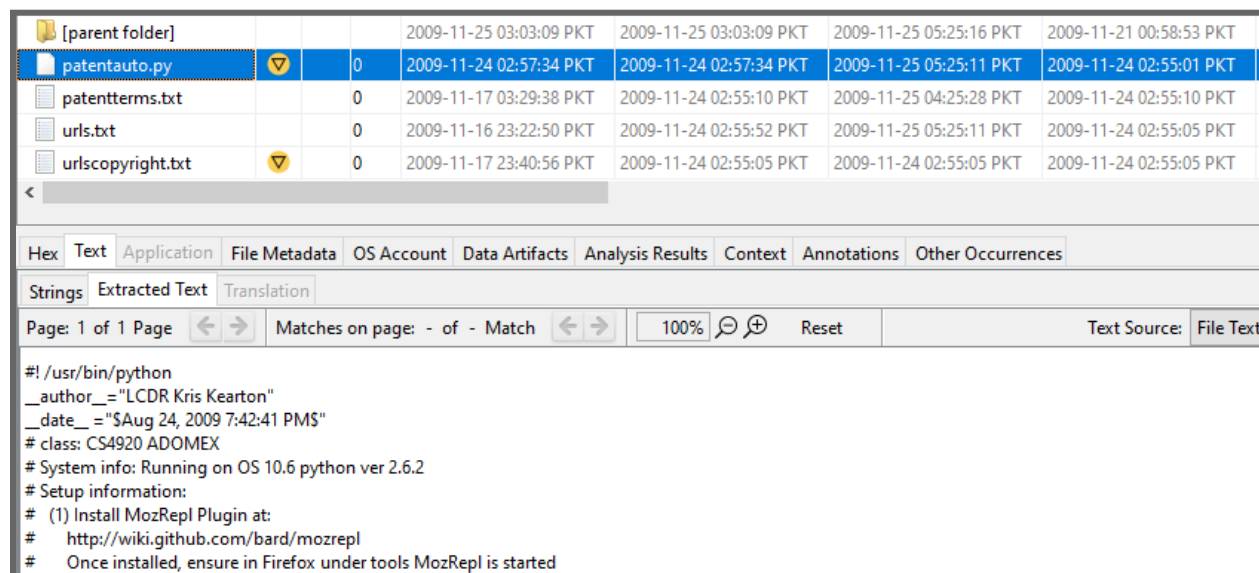
△ Name	S	C	O	Modified Time	Change Time	Access Time	Created Time
E-mail.Ink			0	2009-11-23 21:57:07 PKT	2009-11-23 21:57:07 PKT	2009-11-25 03:02:11 PKT	2009-11-23 21:57:07 PKT
My Computer.Ink			0	2009-11-25 03:03:09 PKT	2009-11-25 03:03:09 PKT	2009-11-25 03:03:09 PKT	2009-11-25 03:03:09 PKT
Pics				2009-11-25 03:16:18 PKT	2009-11-25 03:16:18 PKT	2009-11-25 05:25:16 PKT	2009-11-23 23:24:38 PKT
[current folder]				2009-11-25 03:03:09 PKT	2009-11-25 03:03:09 PKT	2009-11-25 05:25:16 PKT	2009-11-21 00:58:53 PKT
[parent folder]				2009-11-24 02:55:48 PKT	2009-11-24 02:55:48 PKT	2009-11-25 05:23:57 PKT	2009-11-21 00:58:49 PKT
web				2009-11-25 05:26:06 PKT	2009-11-25 05:26:06 PKT	2009-11-25 05:26:06 PKT	2009-11-24 02:54:54 PKT

Desktop folder showing web/, Pics/, etc.

3.13 The Desktop\web Folder — The Smoking Gun

File	Modified	Size
patentauto.py	2009-11-23 21:57:34	3,674 B
urlspatents.txt	2009-11-17 18:40:56	5.37 MB
urlstime_machine.txt	2009-11-16 18:22:50	1.54 MB
urlscopyright.txt	2009-11-17 18:40:56	376 KB
urls.txt	2009-11-16 18:22:50	300 KB
patentterms.txt	2009-11-16 22:29:38	140 B
HighQuality/ and Videos/	(deleted)	unallocated

[patentauto.py](#) header shows it was originally written by "LCDR Kris Kearton" for course CS4920 ADOMEX (2009-08-24) on macOS 10.6, redeployed by Jo on her XP machine. The script drives Firefox via MozRepl to bulk-harvest patent records.



File	Size	Modified
[parent folder]		2009-11-25 03:03:09 PKT
patentauto.py	3,674 B	2009-11-24 02:57:34 PKT
patentterms.txt	140 B	2009-11-17 03:29:38 PKT
urls.txt	300 KB	2009-11-16 23:22:50 PKT
urlscopyright.txt	376 KB	2009-11-17 23:40:56 PKT

Strings | Extracted Text | Translation

Page: 1 of 1 Page | Matches on page: - of - Match | 100% | Reset | Text Source: File Text

```
#!/usr/bin/python
__author__ = "LCDR Kris Kearton"
__date__ = "$Aug 24, 2009 7:42:41 PMS"
# class: CS4920 ADOMEX
# System info: Running on OS 10.6 python ver 2.6.2
# Setup information:
# (1) Install MozRepl Plugin at:
# http://wiki.github.com/bard/mozrepl
# Once installed, ensure in Firefox under tools MozRepl is started
```

Desktop\web folder listing with [patentauto.py](#) preview

3.14 Email Triage

Only 2 emails extracted (msg_25.txt from 2001, msg_43.txt from 2004) — both predate the [M57.biz](#) install. No real email store on disk. Jo used webmail, which is consistent with 13,472 Communication Accounts auto-extracted from Web History.

Source Name	S	C	O	E-Mail From	E-Mail To
 msg_43.txt				; >;	webmaster@python.org;
 msg_25.txt				MAILER-DAEMON@zinfandel.lacita.com;	linuxuser-admin@www.linux.org.uk;

Autopsy E-Mail Messages panel

4. Part B — Memory Forensics (Days 83–85)

4.1 Why Two Frameworks

Volatility 3's windows.netstat throws NotImplementedException on XP, and windows.connscan doesn't exist in V3 at all. So V3 was used wherever possible, with Volatility 2.6 layered in specifically for connections, connscan, cmdline, cmdscan, and consoles.

4.2 Volatility 3 — windows.info

Copy

```
vol.exe -f jo-2009-11-16.mddramimage windows.info
```

Value	Result
NtBuildLab	2600.xpsp_sp3_gdr.090804-1435
CSDVersion	3 (SP3)
Is64Bit	False
KeNumberProcessors	1
SystemTime	2009-11-17 00:22:31 UTC
NtSystemRoot	C:\WINDOWS

All values match the SOFTWARE hive baseline → confirms RAM and disk are from the same machine.

```

Variable      Value
Kernel Base   0x804d7000
DTB           0x39000
Symbols file:  file:///C:/Forensics/Week12/symbols/windows/ntoskrnl.pdb/1B2D0DFE2FB942758D615C901BE04692-2.json.xz
Is64Bit       False
IsPAE         False
layer_name     0 WindowsIntel
memory_layer   1 FileLayer
KdDebuggerDataBlock 0x8054cde0
NTBuildLab     2600.xpsp_sp3_gdr.090804-1435
CSDVersion     3
KdVersionBlock 0x8054cdb8
Major/Minor    15.2600
MachineType    332
KeNumberProcessors 1
SystemTime     2009-11-17 00:22:31+00:00
NtSystemRoot   C:\WINDOWS
NtProductType  NtProductWinNt
NtMajorVersion 5
NtMinorVersion 1
PE MajorOperatingSystemVersion 5
PE MinorOperatingSystemVersion 1
PE Machine     332
PE TimeDateStamp Tue Aug 4 15:14:34 2009

```

Volatility 3 [windows.info](#) output

4.3 Volatility 3 — windows.pslist

47 processes total. Key entries:

PID	PPID	Process	CreateTime (UTC)	Significance
472	416	explorer.exe	2009-11-13 04:49:12	User shell
1932	996	AVGIDSAgent.exe	2009-11-13 04:49:06	Holds SMB to 192.168.1.104:139
3124	472	firefox.exe	2009-11-16 18:47:24	User-launched
3996	472	firefox.exe	2009-11-16 19:34:03	Holds all HTTP connections
2392	3656	soffice.bin	2009-11-16 18:46:13	OpenOffice
3096	472	cmd.exe	2009-11-17 00:21:37	Where mdd was launched
3700	3096	mdd_1.3.exe	2009-11-17 00:22:29	Memory dump tool, alive at capture

Five cmd.exe instances spawned by soffice.bin in a 5-second window — anomalous, suggests OpenOffice macro execution.

```
C:\Forensics\Week12>vol.exe.exe -f jo-2009-11-16.mddramimage windows.pslist
Volatility 3 Framework 2.28.0
Progress: 100.00
PDB scanning finished
```

PID	PPID	ImageFileName	Offset(V)	Threads	Handles	SessionId	Wow64	CreateTime	ExitTime	File outp
4	0	System	0x82fc89c8	58	526	N/A	False	N/A	N/A	Disabled
880	4	smss.exe	0x82e067c8	3	19	N/A	False	2009-11-13 04:48:58.000000 UTC	N/A	Disabled
928	880	csrss.exe	0x82d29a58	13	664	0	False	2009-11-13 04:49:00.000000 UTC	N/A	Disabled
952	880	winlogon.exe	0x82d2f208	18	643	0	False	2009-11-13 04:49:00.000000 UTC	N/A	Disabled
996	952	services.exe	0x82e24d10	16	358	0	False	2009-11-13 04:49:01.000000 UTC	N/A	Disabled
1008	952	lsass.exe	0x82e0bda0	20	353	0	False	2009-11-13 04:49:01.000000 UTC	N/A	Disabled
1180	996	svchost.exe	0x82cd7020	18	199	0	False	2009-11-13 04:49:03.000000 UTC	N/A	Disabled
1268	996	svchost.exe	0x82dff1f8	9	289	0	False	2009-11-13 04:49:03.000000 UTC	N/A	Disabled
1392	996	svchost.exe	0x8297fc08	76	1582	0	False	2009-11-13 04:49:04.000000 UTC	N/A	Disabled
1452	996	svchost.exe	0x82e333a0	5	80	0	False	2009-11-13 04:49:04.000000 UTC	N/A	Disabled
1632	952	avgchsvx.exe	0x82d1ada0	23	280	0	False	2009-11-13 04:49:04.000000 UTC	N/A	Disabled
1640	952	avgrsx.exe	0x82cf7c00	28	243	0	False	2009-11-13 04:49:04.000000 UTC	N/A	Disabled
1756	996	svchost.exe	0x82e2fb10	13	176	0	False	2009-11-13 04:49:05.000000 UTC	N/A	Disabled
1816	1640	avgcsrpx.exe	0x82d32be0	8	167	0	False	2009-11-13 04:49:06.000000 UTC	N/A	Disabled
1904	996	spoolsv.exe	0x82d6fda0	10	106	0	False	2009-11-13 04:49:06.000000 UTC	N/A	Disabled
1932	996	AVGIDSAgent.exe	0x82d383d8	30	525	0	False	2009-11-13 04:49:06.000000 UTC	N/A	Disabled
472	416	explorer.exe	0x82cf36e8	12	531	0	False	2009-11-13 04:49:12.000000 UTC	N/A	Disabled
744	472	avgtray.exe	0x82cc0860	18	313	0	False	2009-11-13 04:49:18.000000 UTC	N/A	Disabled
188	472	hkcmd.exe	0x82ef4910	2	100	0	False	2009-11-13 04:49:19.000000 UTC	N/A	Disabled
856	472	jusched.exe	0x82ccfb28	1	40	0	False	2009-11-13 04:49:20.000000 UTC	N/A	Disabled
900	472	ctfmon.exe	0x829156c8	1	75	0	False	2009-11-13 04:49:20.000000 UTC	N/A	Disabled
924	472	msmsgs.exe	0x82daf758	3	200	0	False	2009-11-13 04:49:21.000000 UTC	N/A	Disabled

pslist output (top)

924	472	msmsgs.exe	0x82daf758	3	200	0	False	2009-11-13 04:49:21.000000 UTC	N/A	Disabled
196	744	AVGIDSMonitor.e	0x828ffbc0	3	37	0	False	2009-11-13 04:49:26.000000 UTC	N/A	Disabled
788	996	svchost.exe	0x82936600	5	107	0	False	2009-11-13 04:49:28.000000 UTC	N/A	Disabled
932	996	avgwdsvc.exe	0x828f8bc0	44	1481	0	False	2009-11-13 04:49:28.000000 UTC	N/A	Disabled
896	996	avgfws0.exe	0x828266a0	30	861	0	False	2009-11-13 04:49:28.000000 UTC	N/A	Disabled
1384	996	jqs.exe	0x82804bc0	5	120	0	False	2009-11-13 04:49:33.000000 UTC	N/A	Disabled
2376	996	avgemc.exe	0x82cc34b8	21	599	0	False	2009-11-13 04:49:38.000000 UTC	N/A	Disabled
2476	932	avgam.exe	0x827a1da0	15	293	0	False	2009-11-13 04:49:40.000000 UTC	N/A	Disabled
2540	932	avgnsx.exe	0x827919e8	25	450	0	False	2009-11-13 04:49:40.000000 UTC	N/A	Disabled
2656	2376	avgcsrpx.exe	0x82783968	3	115	0	False	2009-11-13 04:49:41.000000 UTC	N/A	Disabled
3976	996	alg.exe	0x827eebc0	5	104	0	False	2009-11-13 04:49:54.000000 UTC	N/A	Disabled
1320	2540	avgcsrpx.exe	0x828b7da0	7	174	0	False	2009-11-13 04:55:49.000000 UTC	N/A	Disabled
3180	932	avgcmgr.exe	0x82852020	0	0	0	False	2009-11-16 05:31:50.000000 UTC	2009-11-16 05:31:53.000000 UTC	Disabled
3656	472	soffice.exe	0x82909020	1	20	0	False	2009-11-16 18:46:13.000000 UTC	N/A	Disabled
2392	3656	soffice.bin	0x8290b020	21	554	0	False	2009-11-16 18:46:13.000000 UTC	N/A	Disabled
3860	2392	cmd.exe	0x826d45c0	0	0	0	False	2009-11-16 18:47:16.000000 UTC	2009-11-16 18:47:20.000000 UTC	Disabled
7278	2392	cmd.exe	0x82f20da0	0	0	0	False	2009-11-16 18:47:21.000000 UTC	2009-11-16 18:47:21.000000 UTC	Disabled
3968	2392	cmd.exe	0x826777a8	0	0	0	False	2009-11-16 18:47:21.000000 UTC	2009-11-16 18:47:22.000000 UTC	Disabled
1188	2392	cmd.exe	0x8295fda0	0	0	0	False	2009-11-16 18:47:22.000000 UTC	2009-11-16 18:47:22.000000 UTC	Disabled
3328	2392	cmd.exe	0x82cbbda0	0	0	0	False	2009-11-16 18:47:22.000000 UTC	2009-11-16 18:47:23.000000 UTC	Disabled
3124	472	firefox.exe	0x826be020	0	0	0	False	2009-11-16 18:47:24.000000 UTC	2009-11-16 19:32:51.000000 UTC	Disabled
3996	472	firefox.exe	0x826ddda0	0	0	0	False	2009-11-16 19:34:03.000000 UTC	2009-11-17 00:10:18.000000 UTC	Disabled
480	932	avgsmrpx.exe	0x826994b0	0	0	0	False	2009-11-16 20:03:34.000000 UTC	2009-11-16 20:03:34.000000 UTC	Disabled
3684	932	avgcmgr.exe	0x82623da0	0	0	0	False	2009-11-16 23:47:20.000000 UTC	2009-11-16 23:47:20.000000 UTC	Disabled
2148	3344	soffice.bin	0x82648da0	0	0	0	False	2009-11-17 00:08:10.000000 UTC	2009-11-17 00:08:16.000000 UTC	Disabled
3096	472	cmd.exe	0x82735020	1	33	0	False	2009-11-17 00:21:37.000000 UTC	N/A	Disabled
3700	3096	mdd_1.3.exe	0x82f045d0	1	24	0	False	2009-11-17 00:22:29.000000 UTC	N/A	Disabled

pslist output (full)

4.4 Volatility 3 — windows.pstree

Process tree confirms:

- explorer.exe (472) → cmd.exe (3096) → mdd_1.3.exe (3700) with full command:
mdd_1.3.exe -o joRam01.dmp - All user-launched processes root at explorer.exe — no orphan processes, no anti-forking tricks.

```
WINDOWS\system32\cmd.exe - - -
*** 3860 2392 cmd.exe 0x826d45c0 0 - 0 False 2009-11-16 18:47:16.000000 UTC 2009-11-16 18:47:20.000000 UTC \Device\HarddiskVolume1\
WINDOWS\system32\cmd.exe - - -
* 3996 472 firefox.exe 0x826ddda0 0 - 0 False 2009-11-16 19:34:03.000000 UTC 2009-11-17 00:10:18.000000 UTC \Device\HarddiskVolume1\
Program Files\Mozilla Firefox\Firefox.exe - - -
* 3124 472 firefox.exe 0x826be020 0 - 0 False 2009-11-16 18:47:24.000000 UTC 2009-11-16 19:32:51.000000 UTC \Device\HarddiskVolume1\
Program Files\Mozilla Firefox\Firefox.exe - - -
* 856 472 jusched.exe 0x82ccfb28 1 40 0 False 2009-11-13 04:49:20.000000 UTC N/A \Device\HarddiskVolume1\Program Files\Java\jre6\
bin\jusched.exe "C:\Program Files\Java\jre6\bin\jusched.exe" C:\Program Files\Java\jre6\bin\jusched.exe
* 3096 472 cmd.exe 0x82735020 1 33 0 False 2009-11-17 00:21:37.000000 UTC N/A \Device\HarddiskVolume1\WINDOWS\system32\cmd.exe
"C:\WINDOWS\system32\cmd.exe" C:\WINDOWS\system32\cmd.exe
** 3700 3096 mdd_1.3.exe 0x82f045d0 1 24 0 False 2009-11-17 00:22:29.000000 UTC N/A \Device\HarddiskVolume3\RAM\mdd_1.3.exe mdd_1.3.
exe -o joRam01.dmp E:\RAM\mdd_1.3.exe
* 188 472 hkcmd.exe 0x82ef4910 2 100 0 False 2009-11-13 04:49:19.000000 UTC N/A \Device\HarddiskVolume1\WINDOWS\system32\hkcmd.e
xe "C:\WINDOWS\system32\hkcmd.exe" C:\WINDOWS\system32\hkcmd.exe
2148 3344 soffice.bin 0x82648da0 0 - 0 False 2009-11-17 00:08:10.000000 UTC 2009-11-17 00:08:16.000000 UTC \Device\HarddiskVolume1\
Program Files\OpenOffice.org 3\program\soffice.bin
```

pstree output showing explorer → cmd → mdd chain

4.5 Volatility 2 — imageinfo

Copy

Suggested Profile(s): WinXPSP2x86, WinXPSP3x86

Image date and time: 2009-11-17 00:22:31 UTC+0000

Image local date and time: 2009-11-16 16:22:31 -0800

Profile used: WinXPSP2x86. Local time confirms UTC-8 (PST).

```
C:\Users\Ubaid>cd C:\Forensics\Week12

C:\Forensics\Week12>vol2.exe.exe -f jo-2009-11-16.mddramimage imageinfo
Volatility Foundation Volatility Framework 2.6
INFO      : volatility.debug      : Determining profile based on KDBG search...
           Suggested Profile(s) : WinXPSP2x86, WinXPSP3x86 (Instantiated with WinXPSP2x86)
           AS Layer1            : IA32PagedMemory (Kernel AS)
           AS Layer2            : FileAddressSpace (C:\Forensics\Week12\jo-2009-11-16.mddramimage)
           PAE type             : No PAE
           DTB                  : 0x39000L
           KDBG                 : 0x8054cde0L
           Number of Processors : 1
           Image Type (Service Pack) : 3
           KPCR for CPU 0       : 0xffdff000L
           KUSER_SHARED_DATA    : 0xffdf0000L
           Image date and time  : 2009-11-17 00:22:31 UTC+0000
           Image local date and time : 2009-11-16 16:22:31 -0800
```

Volatility 2 imageinfo output

4.6 Volatility 2 — connscan (Network)

After filtering artefact rows (impossible PIDs / IPs), three connection clusters survived:

Cluster A — Firefox browsing (PID 3996): Google, CNN/Turner, Akamai, Adobe Omniture, Level3, CENIC. All port 80 HTTP. Normal browsing.

Cluster B — SMB session (PID 1932 AVGIDSAgent.exe):

Local	Remote	PID
192.168.1.102:4283	192.168.1.104:139	1932
192.168.1.102:4265	192.168.1.104:139	1932
192.168.1.102:4283	192.168.1.104:139	1932

Port 139 = NetBIOS Session Service / SMB. This is the second exfiltration channel.

Cluster C — Loopback (PID 1384 jqs.exe): Java Quick Starter internal pipe.

0x02656e68	128.201.97.130:0	77.109.83.100:24576	144
0x02657e68	0.0.0.0:0	0.0.0.0:0	0
0x02659008	192.168.1.102:4244	157.166.224.107:80	3996
0x0265c270	192.168.1.102:4283	192.168.1.104:139	1932
0x0267d600	0.0.0.0:24706	1.0.0.0:2080	0
0x02693b38	192.168.1.102:4190	198.189.255.76:80	3996
0x026b1d00	192.168.1.102:4412	198.189.255.76:80	3996

connscan full output

4.7 WHOIS of Remote IPs

IP	Org	Location
74.125.19.156 / .148	Google	Mountain View, CA
157.166.x.x	Turner Broadcasting (CNN)	Atlanta, GA
66.235.142.20	Adobe (Omniure)	Lehi, UT
72.247.247.9	Akamai	Cambridge, MA
198.189.255.76 / .81	CENIC	Cypress, CA
4.23.63.126 / 199.93.34.126	Level 3 (Lumen)	various

All Cluster A IPs = normal commercial web destinations.

4.8 Volatility 2 — cmdline / cmdscan / consoles ★ Most Important Artefact

cmdscan recovered the actual three commands typed into cmd.exe (PID 3096):

Copy

Cmd #0: e:

Cmd #1: cd RAM

Cmd #2: mdd_1.3.exe -o joRam01.dmp

This proves:

- Jo switched to drive E: (a USB drive)
- Navigated into a folder named RAM
- Ran ManTech's memory dump tool herself, writing joRam01.dmp

consoles also recovered the screen output: "ManTech Physical Memory Dump Utility... Dumping 765.99 MB of physical memory to file joRam01.dmp." — matches the size of our RAM image exactly.

```
avgsrcmax.exe pid: 480
*****
avgcmgr.exe pid: 3684
*****
soffice.bin pid: 2148
*****
cmd.exe pid: 3096
Command line : "C:\WINDOWS\system32\cmd.exe"
*****
mdd_1.3.exe pid: 3700
Command line : mdd_1.3.exe -o joRam01.dmp
```

cmdline output for mdd

```

CommandProcess: csrss.exe Pid: 928
CommandHistory: 0x4f4660 Application: cmd.exe Flags: Allocated, Reset
CommandCount: 3 LastAdded: 2 LastDisplayed: 2
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x768
Cmd #0 @ 0x4f1f90: e:
Cmd #1 @ 0x10c8f98: cd RAM
Cmd #2 @ 0x4f4820: mdd_1.3.exe -o joRam01.dmp
*****
CommandProcess: csrss.exe Pid: 928
CommandHistory: 0x4f6628 Application: mdd_1.3.exe Flags: Allocated
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x45c
*****
CommandProcess: csrss.exe Pid: 928
CommandHistory: 0x10c86f8 Application: avgidsmonitor.exe Flags: Allocated
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x594

```

cmdscan output with the 3 commands

```

E:\RAM>mdd_1.3.exe -o joRam01.dmp
-> mdd
-> ManTech Physical Memory Dump Utility
    Copyright (C) 2008 ManTech Security & Mission Assurance

-> This program comes with ABSOLUTELY NO WARRANTY; for details use option '-w'
    This is free software, and you are welcome to redistribute it
    under certain conditions; use option '-c' for details.

-> Dumping 765.99 MB of physical memory to file 'joRam01.dmp'.

```

console screen output

4.9 Volatility 3 — windows.filescan

Filtered for patent / m57 / .doc / .xls / usb / removable. Notable hits:

- /jo/my documents/downloads/schema.txt — in RAM but NOT on disk → deleted between 2009-11-17 and 2009-11-24 (counter-forensic activity)
- /windows/system32/drivers/usbstor.sys — USB driver live in kernel memory
- /jo/local settings/history/history.ie5/index.dat — IE history cached in RAM

0x270ebe8	華國菜uments and Settings\All Users\Application Data\avg9\Log\privlog.cfg
0x270f340	\Program Files\OpenOffice.org 3\Basis\share\registry\schema\org\openoffice\
0x2710258	\Documents and Settings\All Users\Application Data\avg9\Antispam\scoffset.b
0x27103c8	\9669dca9-0326-4e43-9d64-3d12a2394e71

filesan filtered output in Notepad

4.10 Volatility 3 — windows.registry.userassist

UserAssist UEME_RUNPATH last-write: 2009-11-17 00:21:37 UTC — exactly matches cmd.exe (PID 3096) CreateTime. Independent confirmation that Jo launched cmd.exe via the shell.

```
05 00 00 00 06 00 00 00 30 8c 06 b3 f2 66 ca 01 .....0....f..
* 0xe2988b60 \Device\HarddiskVolume1\Documents and Settings\Jo\NTUSER.DAT NTUSER.DAT\Software\Microsoft\Windows\Cu
11D0-9888-006097DEACF9}\Count 2009-11-17 00:21:37.000000 UTC Value
    UEME_RUNPATH:C:\WINDOWS\system32\cmd.exe 5 1 N/A N/A 2009-11-17 00:21:37.000000 UTC
05 00 00 00 06 00 00 00 80 be e9 ec 1b 67 ca 01 .....g..WARNING volatility3.plugins.windows.registry.userassist
lume1\Documents and Settings\LocalService\NTUSER.DAT at 0xe15aa380
WARNING volatility3.plugins.windows.registry.userassist: UserAssist key not found in \Device\HarddiskVolume1\Documents
```

userassist output

4.11 RAM-to-Disk Correlation (Key Rows)

RAM Finding	Disk Corroboration	Verdict
PID 1932 → SMB to 192.168.1.104:139	NTUSER.DAT \Network\Z mapped to \\192.168.1.1\m57 as m57admin	✓ Confirmed
mdd_1.3.exe -o joRam01.dmp	No mdd_1.3.exe on C: drive — tool was on E: USB	✓ Confirmed counter-forensic
schema.txt in RAM	Not on disk	✓ Confirmed deletion
UserAssist write 2009-11-17 00:21:37	Matches cmd.exe CreateTime	✓ Confirmed user launch

5. Part C — Exfiltration and Correlation (Days 86–88)

5.1 USB Identification — Which One Was the Crime USB?

Favourites USB: HighQuality/ (84 files), Videos/ (14 files) — consumer media only. macOS Spotlight metadata present.

Work USB: Papers1/ through Papers17/ (17 × 96 = 1,632 files), XFER-11-20-2010/ staging folder, deleted _SC00013.JPG–_SC00019.JPG screenshots. Labelled "work" by Jo.

Verdict: Work USB is the criminal device.

jo-favorites-usb-2009-12-11.E01

vol1 (Unallocated: 0-62)

vol2 (Win95 FAT32 (0x0b): 63-2047940)

\$OrphanFiles (0)

\$Unalloc (1)

.fsevents (0)

.Spotlight-V100 (3)

.Trashes (6)

_NTITL~1 (0)

HighQuality (84)

Videos (14)

vol3 (Unallocated: 2047941-2047999)

jo-work-usb-2009-12-11.E01_107610 Host

File Viewer

Name	S	C	O	Modified Time
DSC00000.JPG			0	2009-11-12 14:11:28 PKT
DSC00051.JPG			0	2009-11-12 14:11:32 PKT
HighQuality				2009-11-17 16:41:40 PKT
JO FAVORITE (Volume Label Entry)				2009-11-20 14:30:34 PKT
KittyMontage.mov				2009-11-20 14:18:56 PKT
MontereyKitty.m4v				2009-11-20 13:57:26 PKT
MontereyKittyHQ.m4v				2009-11-20 14:00:00 PKT
TiggerTheCat.m4v				2009-11-18 21:50:54 PKT
Videos				2009-11-20 14:25:58 PKT
_NTITL~1				2009-11-20 14:25:58 PKT

Favourites USB volume structure

jo-work-usb-2009-12-11.E01

vol1 (Unallocated: 0-62)

vol2 (Win95 FAT32 (0x0b): 63-255968)

\$OrphanFiles (0)

\$Unalloc (1)

.fsevents (4)

.fsevents (0)

.Spotlight-V100 (3)

.Trashes (4)

HighQuality (0)

Papers1 (96)

Papers10 (96)

Papers11 (96)

Papers12 (96)

Papers13 (96)

Papers14 (96)

Papers15 (96)

Papers16 (96)

Papers17 (96)

Papers2 (96)

Papers3 (96)

Papers4 (96)

Papers5 (96)

Papers6 (96)

Papers7 (96)

Papers8 (96)

Papers9 (96)

XFER-11-20-2010 (8)

vol3 (Unallocated: 255969-255999)

Name	S	C	O	Modified Time
SFAT1			0	0000-00-00 00:00:00
SFAT2			0	0000-00-00 00:00:00
SMBR			0	0000-00-00 00:00:00
\$OrphanFiles				0000-00-00 00:00:00
\$Unalloc				0000-00-00 00:00:00
.Spotlight-V100				2009-11-20 18:26:12 PKT
.Trashes				2009-11-20 18:26:12 PKT
._Trashes			0	2009-11-20 18:26:12 PKT
._Cat.m4v				2009-11-20 18:35:34 PKT
._Cat.m4v				2009-11-20 18:31:48 PKT

Hex

Text

Application

File Metadata

OS Account

Data Artifacts

Analysis Results

Work USB volume structure showing Papers1-17

5.2 Work USB Key Files

File	Status	Notes
Papers1 to Papers17 (1,632 files)	Allocated	Bulk patent corpus
XFER-11-20-2010/	Allocated	Dated staging folder
_SC00013.JPG – _SC00019.JPG	Unallocated	7 screen captures, ~60 KB each, 2009-11-20 21:36–21:56 UTC
_TigerTheCat copy.m4v	Carved	MD5 edeca2340b4125f4ef4c949f9190ba5c — matches macOS resource fork on Jo's disk
TiggerTheCat copy.m4v	Allocated	MD5 9bd2dd0f202046eb032bdf46ba91fefa

	_SC00013.JPG			2009-11-11 14:18:10 PKT	0000-00-00 00:00:00	2009-11-20 05:00:00 PKT
	_SC00014.JPG			2009-11-11 14:18:16 PKT	0000-00-00 00:00:00	2009-11-20 05:00:00 PKT
< [Progress Bar]						

Hex	Text	Application	File Metadata	OS Account	Data Artifacts	Analysis Results	Context	Annotations	Other Occurrences
-----	------	-------------	---------------	------------	----------------	------------------	---------	-------------	-------------------

Metadata

Name:

/img_jo-work-usb-2009-12-11.E01/vol_vol2/_SC00013.JPG

Type:

File System

MIME Type:

application/octet-stream

Size:

60285

File Name Allocation:

Unallocated

Metadata Allocation:

Unallocated

Modified:

2009-11-11 14:18:10 PKT

Accessed:

2009-11-20 05:00:00 PKT

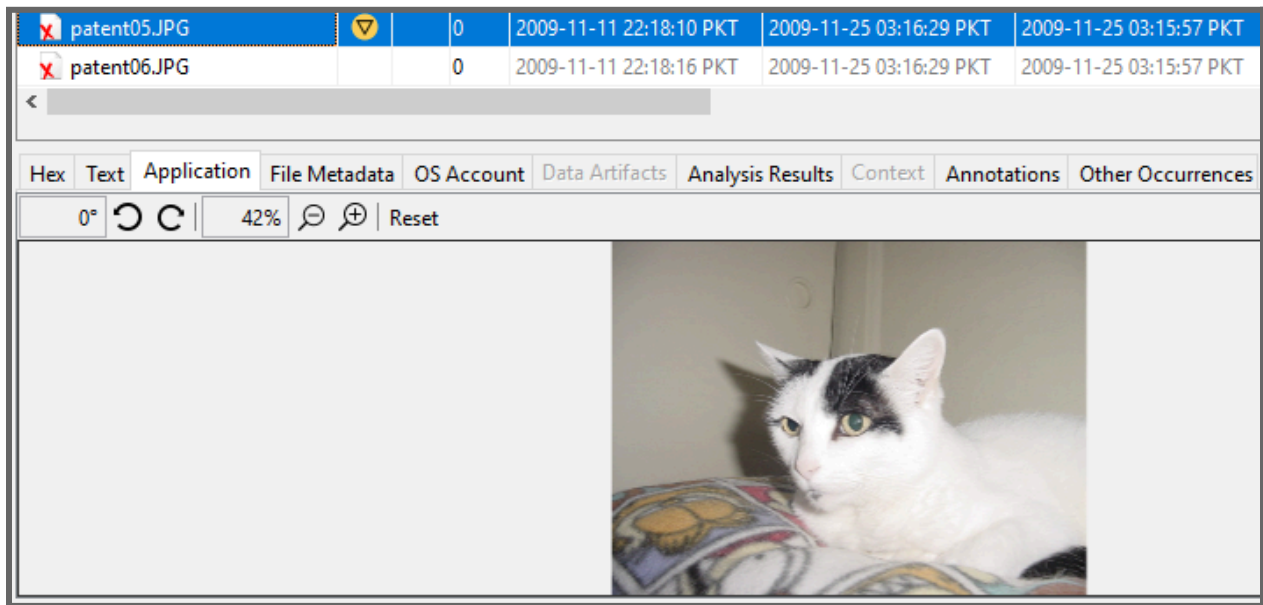
Created:

2009-11-21 02:36:24 PKT

Changed:

0000-00-00 00:00:00

Work USB file listing with _SC00013.JPG metadata



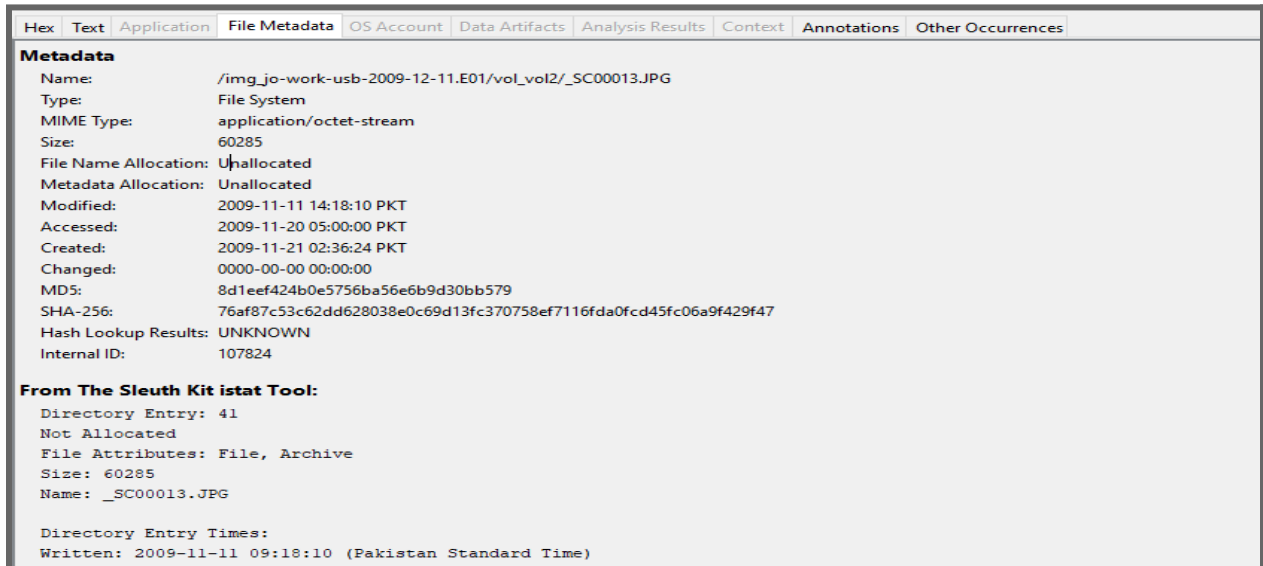
Pics folder on primary disk with patent JPEGs

5.3 Hash Match Proof

The macOS resource-fork hash matches between disk and Work USB → proves the USB transited Jo's macOS home machine.

5.4 Deleted File Recovery — Work USB

8 files recovered, totalling 374 KB. Confirms FAT32 cluster data wasn't overwritten before USB seizure.

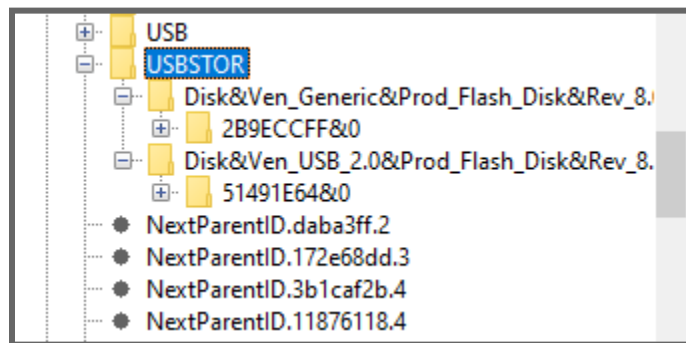


Autopsy metadata for recovered deleted file

5.5 USBSTOR Registry Evidence (SYSTEM hive)

ControlSet001\Enum\USBSTOR:

Sub-key	FriendlyName	Last Write (UTC)
Disk&Ven_USB_2.0&Prod_Flash_Disk\5149E64&0	USB 2.0 Flash Disk	2009-11-23 21:54:45
Disk&Ven_Generic&Prod_Flash_Disk\2B9EC CFF&0	Generic Flash Disk	2009-11-24 22:01:54



USBSTOR key with both devices

Metadata		
Name: 2B9ECCFF&0		
Number of subkeys: 2		
Number of values: 13		
Modification Time: 2009-11-24 22:01:54 GMT+00:00		
Values		
Name	Type	Value
DeviceDesc	REG_SZ	Disk drive
Capabilities	REG_DWORD	0x00000010 (16)
UINumber	REG_DWORD	0x00000000 (0)
HardwareID	REG_MULTI_SZ	USBSTOR\DiskGeneric_Flash_Disk____8.00, USB...
CompatibleIDs	REG_MULTI_SZ	USBSTOR\Disk, USBSTOR\RAW,
ClassGUID	REG_SZ	{4D36E967-E325-11CE-BFC1-08002BE10318}
Service	REG_SZ	disk
ConfigFlags	REG_DWORD	0x00000000 (0)
ParentIdPrefix	REG_SZ	78&1390c1d1&0
Driver	REG_SZ	{4D36E967-E325-11CE-BFC1-08002BE10318}\0002
Class	REG_SZ	DiskDrive
Mfg	REG_SZ	(Standard disk drives)
FriendlyName	REG_SZ	Generic Flash Disk USB Device

Generic Flash Disk sub-key detail

Metadata		
Name: 51491E64&0		
Number of subkeys: 2		
Number of values: 13		
Modification Time: 2009-11-23 21:54:45 GMT+00:00		
Values		
Name	Type	Value
DeviceDesc	REG_SZ	Disk drive
Capabilities	REG_DWORD	0x00000010 (16)
UINumber	REG_DWORD	0x00000000 (0)
HardwareID	REG_MULTI_SZ	USBSTOR\DiskUSB_2.0_Flash_Disk____8.00, USB...
CompatibleIDs	REG_MULTI_SZ	USBSTOR\Disk, USBSTOR\RAW,
ClassGUID	REG_SZ	{4D36E967-E325-11CE-BFC1-08002BE10318}
Service	REG_SZ	disk
ConfigFlags	REG_DWORD	0x00000000 (0)
ParentIdPrefix	REG_SZ	7&18ec28b&0
Driver	REG_SZ	{4D36E967-E325-11CE-BFC1-08002BE10318}\0001
Class	REG_SZ	DiskDrive
Mfg	REG_SZ	(Standard disk drives)
FriendlyName	REG_SZ	USB 2.0 Flash Disk USB Device

USB 2.0 Flash Disk sub-key detail

5.6 MountPoints2 — Drive Letter Mapping (NTUSER.DAT)

Sub-key	Type	Last Write (UTC)
##192.168.1.1#m57	Network share	2009-11-21 01:00:46
E	Removable USB	2009-11-24 22:01:52
{300cbeed-...} (= ##192.168.1.1#m57)	Network alias	2009-11-23 21:54:50

The {300cbeed} write came 5 seconds after the USB 2.0 Flash Disk was attached → USB and SMB share active in the same session.

Metadata		
Name: ##192.168.1.1#m57		
Number of subkeys: 0		
Number of values: 4		
Modification Time: 2009-11-21 00:06:46 GMT+00:00		
Values		
Name	Type	Value
BaseClass	REG_SZ	Drive
_CommentFromDesktopINI	REG_SZ	(value not set)
_LabelFromDesktopINI	REG_SZ	(value not set)
_AutorunStatus	REG_BIN	01 01 FF FF FF FF FF FF FF FF FF FF FF FF FF...

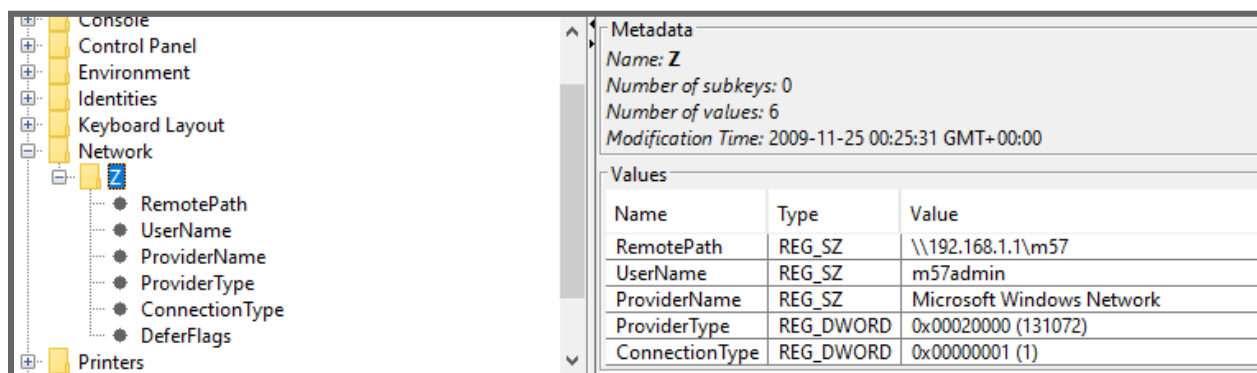
MountPoints2 with ##192.168.1.1#m57 highlighted

5.7 SMB / Network Share Evidence (NTUSER.DAT\Network\Z)

Value	Data
RemotePath	\\192.168.1.1\m57
UserName	m57admin (NOT Jo's own account)
ProviderName	Microsoft Windows Network
ConnectionType	1 (DiskShare)

Jo used a privileged corporate credential (m57admin) to access the M57 share.

Note: RAM connscan shows SMB to 192.168.1.104 , while NTUSER.DAT maps 192.168.1.1 . These are two different LAN hosts — both involved.



NTUSER.DAT \Network\Z key showing m57admin

5.8 Browser History and Web Search

3,373 Web History entries, 14 Web Search entries. Significant searches:

#	Date (UTC)	Term
10	2009-11-23 18:20:00	python (3 min before Python download)
12	2009-11-24 22:08:23	quicktime
13	2009-11-24 22:08:23	http://Quicktime.7-n0w.com ⚠ NOT an Apple domain

Web History rows to patft.uspto.gov/netacgi/nph-Parser between 2009-11-23 22:02:28 and 22:15:37 UTC — live USPTO patent database queries in the same session that patentauto.py was materialized.

Source Name	S	C	O	Domain	Text	Program Name	Date Accessed	Data Source
places.sqlite				google.com	mozrepl	FireFox Analyzer	2009-11-23 23:17:55 PKT	jo-2009-11-24.E01
places.sqlite				google.com	python	FireFox Analyzer	2009-11-23 23:20:00 PKT	jo-2009-11-24.E01
places.sqlite				google.com	cnn	FireFox Analyzer	2009-11-23 23:22:31 PKT	jo-2009-11-24.E01
places.sqlite				google.com	quicktime	FireFox Analyzer	2009-11-25 03:08:23 PKT	jo-2009-11-24.E01
places.sqlite				google.com	http://Quicktime.7-n0w.com	FireFox Analyzer	2009-11-25 03:08:23 PKT	jo-2009-11-24.E01
index.dat				google.com	firefox	Internet Explorer Analyzer	2009-11-20 19:33:47 PKT	jo-2009-11-24.E01
index.dat				google.com	firefox	Internet Explorer Analyzer	2009-11-20 19:33:47 PKT	jo-2009-11-24.E01
index.dat				google.com	firefox	Internet Explorer Analyzer	2009-11-20 19:33:46 PKT	jo-2009-11-24.E01
index.dat				google.com	fire	Internet Explorer Analyzer	2009-11-20 19:33:45 PKT	jo-2009-11-24.E01
index.dat				google.com	firefo	Internet Explorer Analyzer	2009-11-20 19:33:45 PKT	jo-2009-11-24.E01
index.dat				mozilla.com	firefox	Internet Explorer Analyzer	2009-11-20 19:33:57 PKT	jo-2009-11-24.E01
index.dat				google.com	fir	Internet Explorer Analyzer	2009-11-20 19:33:45 PKT	jo-2009-11-24.E01
index.dat				google.com	f	Internet Explorer Analyzer	2009-11-20 19:33:45 PKT	jo-2009-11-24.E01
index.dat				google.com	firefox	Internet Explorer Analyzer	2009-11-20 19:33:46 PKT	jo-2009-11-24.E01

Web Search panel with 14 entries

Name	Keyword Preview	Location	Modifi
Web History Artifact	url : http://patft.uspto.gov/netacgi/nph-parser?	/img_jo-2009-11-24.E01/vol_vol2/Documents and Sett...	2009-11-24
Web History Artifact	url : http://patft.uspto.gov/netacgi/nph-parser?	/img_jo-2009-11-24.E01/vol_vol2/Documents and Sett...	2009-11-24
Web History Artifact	url : http://patft.uspto.gov/netacgi/nph-parser?	/img_jo-2009-11-24.E01/vol_vol2/Documents and Sett...	2009-11-24
Web History Artifact	url : http://patft.uspto.gov/netacgi/nph-parser?	/img_jo-2009-11-24.E01/vol_vol2/Documents and Sett...	2009-11-24
Web History Artifact	url : http://patft.uspto.gov/netacgi/*****	/img_jo-2009-11-24.E01/vol_vol2/Documents and Sett...	2009-11-24
Web History Artifact	url : http://patft.uspto.gov/netacgi/doccrawler%	/img_jo-2009-11-24.E01/vol_vol2/Documents and Sett...	2009-11-24
_CACHE_001_	http://patft.uspto.gov/netacgi/nph-parser?	/img_jo-2009-11-24.E01/vol_vol2/Documents and Sett...	2009-11-24
patentauto.py	#http://patft.uspto.gov/netacgi/nph-parser?	/img_jo-2009-11-24.E01/vol_vol2/Documents and Sett...	2009-11-24
E01FF666d01	&homeurl=http%3a%2f%2fpatft.uspto.gov%2fnetacgi/*****	/img_jo-2009-11-24.E01/vol_vol2/Documents and Sett...	2009-11-24
65830294d01	&homeurl=http%3a%2f%2fpatft.uspto.gov%2fnetacgi/*****	/img_jo-2009-11-24.E01/vol_vol2/Documents and Sett...	2009-11-24

Hex Text Application Source File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Result: 159 of 225 Result

Web History

Visit Details

Title: Not Found

Date Accessed: 2009-11-25 05:10:51 PKT

Domain: uspto.gov

URL: http://patft.uspto.gov/netacgi/*****

Program Name: FireFox Analyzer

Source

Host: jo-2009-11-24.E01_1 Host

Data Source: jo-2009-11-24.E01

Web History with USPTO patft queries

5.9 LNK Files and LECmd

Extracted 13 LNK files from /Jo/Recent/. Parsed with:

Copy

LECmd.exe -d LNK_Files\ --csv LNK_Output\ --dt "yyyy-MM-dd HH:mm:ss" --utc

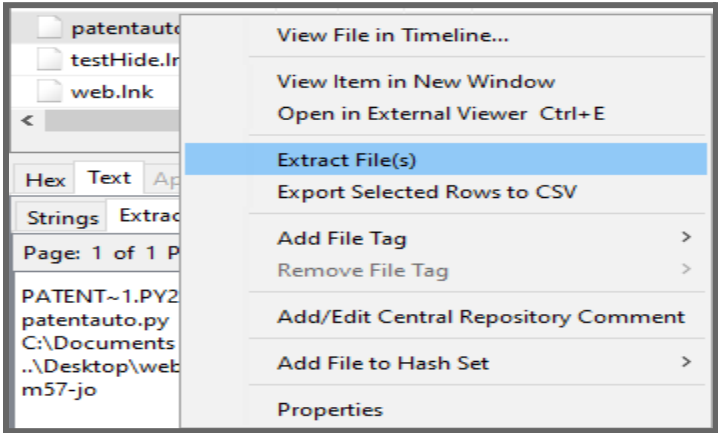
Critical LNKs:

LNK	SourceCreated (UTC)	Target
diit-1.5.lnk	2009-11-23 18:24:43	Java JAR
testHide.lnk	2009-11-23 18:25:53	Hidden subfolder 
patentauto.lnk	2009-11-23 21:57:24	patentauto.py
web.lnk	2009-11-23 21:57:24	Desktop\web
STU.lnk	2009-11-24 22:01:53	STU.doc
HighQuality.lnk	2009-11-24 22:14:45	(deleted ~2 min later)

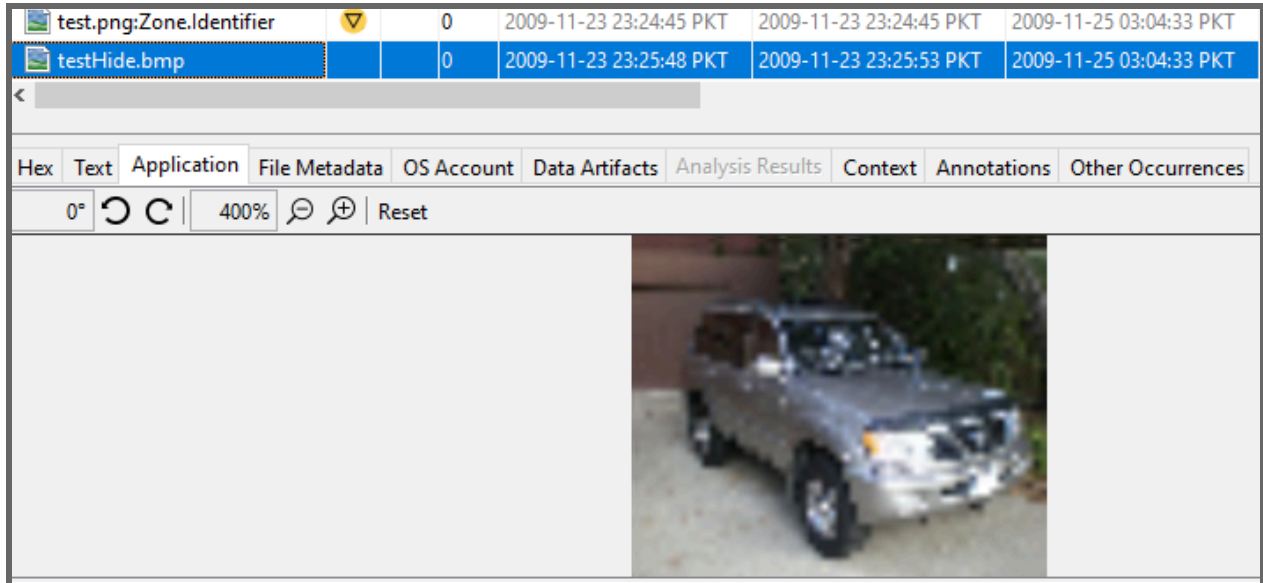
testHide — verb of counter-forensic intent. patentauto.lnk + web.lnk opened in the same minute.

△ Name	S	C	O	Modified Time	Change Time	Access Time
Downloads.lnk			0	2009-11-25 03:01:54 PKT	2009-11-25 03:01:54 PKT	2009-11-25 03:01:54 PKT
HighQuality.lnk			0	2009-11-25 03:14:45 PKT	2009-11-25 03:14:45 PKT	2009-11-25 03:14:45 PKT
Pics.lnk			0	2009-11-23 23:25:53 PKT	2009-11-23 23:25:53 PKT	2009-11-23 23:25:53 PKT
STU.lnk			0	2009-11-25 03:01:53 PKT	2009-11-25 03:01:53 PKT	2009-11-25 03:01:53 PKT
[current folder]				2009-11-25 03:14:45 PKT	2009-11-25 03:14:45 PKT	2009-11-25 03:14:45 PKT
[parent folder]				2009-11-24 02:55:48 PKT	2009-11-24 02:55:48 PKT	2009-11-25 05:23:57 PKT
diit-1.5.lnk			0	2009-11-23 23:24:50 PKT	2009-11-23 23:24:50 PKT	2009-11-23 23:24:50 PKT
patentauto.lnk			0	2009-11-24 02:57:24 PKT	2009-11-24 02:57:24 PKT	2009-11-24 02:57:24 PKT
testHide.lnk			0	2009-11-23 23:25:53 PKT	2009-11-23 23:25:53 PKT	2009-11-23 23:25:53 PKT
web.lnk				No comments found 02:57:24 PKT	2009-11-24 02:57:24 PKT	2009-11-24 02:57:24 PKT

Recent folder listing



Right-click Extract Files menu



Pics folder showing patent JPEGs and testHide.bmp

5.10 Counter-Forensic Activity

- schema.txt — in RAM (2009-11-17) but deleted from disk before 2009-11-24
- HighQuality/ and Videos/ inside Desktop/web — directory entries allocated, contents unallocated. Deleted at 2009-11-24 22:16:29 UTC (minutes before disk seizure)
- testHide.lnk pointing to a Hidden subfolder of Pics
- No bulk archives (.zip/.rar/.7z) — file-by-file copy approach
- No Recycle Bin entries — Shift+Delete was used or bin emptied

5.11 Master Event Sequence (UTC)

Timestamp (UTC)	Src	Event
2009-11-13 04:48:58	DSK	System boot
2009-11-17 00:21:37	RAM	★ Jo launches cmd.exe via shell
2009-11-17 00:22:29	RAM	★ mdd_1.3.exe issued
2009-11-17 00:22:31	RAM	★ MEMORY CAPTURE INSTANT
2009-11-17 (session)	RAM	★ SMB session to 192.168.1.104:139 (PID 1932)

2009-11-17 → 11-24	DEL	★ schema.txt deleted from disk
2009-11-20 14:34:50	DSK	Windows install finalised
2009-11-20 19:35:57	DSK	Jo account created
2009-11-21 01:00:46	NET	★ \\192.168.1.1\m57 mapped Z: as m57admin
2009-11-23 18:20:00	DSK	Google search "python"
2009-11-23 18:23:03	DSK	python-2.6.4.msi downloaded
2009-11-23 18:25:53	DSK	★ testHide.lnk created
2009-11-23 21:54:14	DSK	★ Desktop\web\ folder created
2009-11-23 21:54:45	DSK	★ USB 2.0 Flash Disk attached
2009-11-23 21:54:50	NET	★ M57 share re-accessed (5s after USB)
2009-11-23 21:55:01	DSK	★ patentauto.py materialised
2009-11-23 21:55:05	DSK	★ URL corpus (7.6 MB) materialised
2009-11-23 21:57:24	DSK	★ Jo opens patentauto.py and web folder
2009-11-23 22:02–22:15	DSK	★ Live USPTO patft queries
2009-11-24 01:28:50	DSK	Final login session (count=6)
2009-11-24 22:01:52	DSK	★ E: drive letter assigned
2009-11-24 22:01:53	DSK	STU.doc downloaded and opened

2009-11-24 22:01:54	DSK	★ Generic Flash Disk USB attached
2009-11-24 22:08:23	DSK	★ Search: http://Quicktime.7-n0w.com
2009-11-24 22:08:37	DSK	QuickTimeInstaller.exe downloaded
2009-11-24 22:16:29	DEL	★ HighQuality/Videos folders deleted (counter-forensic)
2009-11-24 (window)	DSK	★ DISK SEIZURE
2009-11-25 05:46:24	DSK	Last hive write
2009-12-11	USB	Both USBs acquired
2010-11-20	USB	(folder name) XFER-11-20-2010 — future-dated staging

6. Part D — Conclusions

6.1 What the Evidence Establishes

Across three evidence sources, nine registry hives, and seven Volatility plugins, one coherent picture emerges. Jo, the registered owner of workstation M57-JO at M57.biz, was running an automated browser-driven patent-harvesting operation in November 2009. The operation used `patentauto.py` (Python + Firefox + MozRepl) with 7.6 MB of URL lists (~50,000+ URLs) staged in `Desktop\web`. Output was exfiltrated through two channels: (a) a labelled "work" USB carrying 17 Papers stacks of 96 files each plus an XFER-11-20-2010 staging folder, and (b) a mapped network share Z: to `\\192.168.1.1\m57` authenticated as `m57admin`. A second LAN host (192.168.1.104:139) was also contacted via SMB. Counter-forensic activity is evident in the deletion of `schema.txt`, the `HighQuality/Videos` subfolders, and the `testHide` LNK pointing to a Hidden directory.

6.2 Question 1 — Jo's Job Designation

PATENT RESEARCHER at M57.biz.

Evidence:

- SOFTWARE hive: RegisteredOrganization=M57.biz, RegisteredOwner=JO
- Hostname M57-JO
- Desktop\web\ contains patentauto.py + 7.6 MB of patent URL lists
- Desktop\Pics\ contains patent05.JPG, patent10.JPG, patent11.JPG, patent14.JPG, patent26.JPG
- Web History contains live USPTO patft.uspto.gov queries
- M57.biz business profile = patent search firm with two patent researchers (Jo and Charlie)

6.3 Question 2 — The Specific Crime

INTELLECTUAL PROPERTY THEFT / CORPORATE ESPIONAGE — unauthorised exfiltration of patent research materials and abuse of privileged m57admin credentials.

Evidence:

- Work USB carries 1,632+ files in Papers1–Papers17 stacks
- XFER-11-20-2010 staging folder with date-embedded name
- USB labelled "work" by Jo
- m57admin credential used (not Jo's jo account) to map \\192.168.1.1\m57
- USB attach and SMB share re-access within 5 seconds
- Counter-forensic deletion: schema.txt, HighQuality/, Videos/, testHide LNK
- Live USPTO queries in same session as URL corpus materialisation

6.4 Question 3 — Exfiltration Mechanism

TWO-CHANNEL PARALLEL OPERATION:

Channel 1 — Removable USB Media:

1. USB 2.0 Flash Disk attached 2009-11-23 21:54:45 UTC
2. patentauto.py + URL corpus materialised in Desktop\web
3. USPTO queries executed 22:02–22:15 UTC
4. Generic Flash Disk attached 2009-11-24 22:01:54 UTC (drive E:)
5. Work USB later found with 17 Papers stacks + staging folder + 7 deleted screenshots
6. macOS Spotlight metadata on both USBs → device also transited Jo's home macOS

Channel 2 — SMB / Network File Share:

1. \\192.168.1.1\m57 mapped as Z: with m57admin on 2009-11-21
2. Share re-accessed 5 seconds after USB attach on 2009-11-23
3. AVGIDSAgent.exe (PID 1932) held SMB sessions to 192.168.1.104:139 during the 2009-11-17 RAM session
4. Aggregation from multiple internal LAN sources before export to USB

6.5 Evidentiary Gaps

- RAM captured 7 days before disk → temporal asymmetry
- USB images acquired 17 days after disk → some USB content may post-date disk
- STU.doc content not parsed
- QuickTimeInstaller.exe hash not compared against Apple's official 7.6.5
- LAN host 192.168.1.104 not attributed
- Pics\Hidden subfolder not exhaustively examined
- The five cmd.exe instances under soffice.bin — OpenOffice macro not identified

6.6 Recommended Next Steps

- Hash QuickTimeInstaller.exe vs Apple's signed distribution
 - Extract and parse STU.doc
 - Identify 192.168.1.104 via DHCP/ARP records from M57 IT
 - Hash-match Papers1–17 files against urlspatents.txt URLs
 - Examine Pics\Hidden subfolder
 - Sweep Web History for webmail / file-sharing services
 - Identify recipient of XFER-11-20-2010 transfer
 - Image Jo's macOS home machine
 - Establish chain of custody for m57admin credential
-

7. Final Summary Table

Question	Answer	Primary Evidence
Job Designation	Patent Researcher	SOFTWARE hive (M57.biz/JO), hostname M57-JO, patentauto.py + 7.6 MB URL corpus, USPTO Web History
Crime	IP Theft / Corporate Espionage	Work USB with 1,632 files in Papers1-17, m57admin privileged access, counter-forensic deletions, testHide LNK
Mechanism	Two-channel: USB + SMB	USBSTOR (2 devices), MountPoints2 (E: + Z: share), RAM connscan PID 1932 → 192.168.1.104:139, NTUSER.DAT\Network\Z

8. Annex A — Evidence File Inventory

File	Type	Size
jo-2009-11-24.E01	EnCase E01	~1.7 GB
jo-2009-11-16.mddramimage	Raw RAM	765.99 MB
jo-favorites-usb-2009-12-11.E01	EnCase E01	216 MB
jo-work-usb-2009-12-11.E01	EnCase E01	112 MB
Registry\SOFTWARE	Extracted hive	12.85 MB
Registry\SYSTEM	Extracted hive	3.41 MB
Registry\NTUSER.DAT (Jo)	Extracted hive	262 KB
Output\pslist.txt	Vol3 plugin	4.7 KB
Output\pstree.txt	Vol3 plugin	10.4 KB
Output\connscan.txt	Vol2 plugin	4.6 KB
Output\cmdline.txt	Vol2 plugin	7.3 KB
Output\cmdscan.txt	Vol2 plugin	0.9 KB
Output\consoles.txt	Vol2 plugin	3.1 KB
Output\filescaan.txt	Vol3 plugin	341 KB
Output\userassist.txt	Vol3 plugin	12.7 KB
LNK_Output*.csv	LECcmd output	~20 KB
Web_History_*.csv	Autopsy export	48.6 KB

9. Annex B — Volatility Plugin Matrix (XP)

Purpose	Vol 3	Vol 2	Used	Notes
OS profile	windows.info ✓	imageinfo ✓	Both	Cross-corroboration
Process list	windows.pslist ✓	pslist ✓	V3	Identical on XP
Process tree	windows.pstree ✓	pstree ✓	V3	
Network (XP)	windows.netstat ✗	connections/ connscan ✓	V2 only	V3 throws NotImplementedError
Cmd args	windows.cmdline ✓	cmdline ✓	V2	Consistency
Cmd history	(none)	cmdscan ✓	V2 only	V3 lacks plugin
Console buffer	(none)	consoles ✓	V2 only	V3 lacks plugin
File scan	windows.filescan ✓	filescan ✓	V3	
UserAssist	windows.registry.us erassist ✓	userassist ✓	V3	

10. Annex C — Hash Inventory (Selected)

Source	File	MD5
Primary disk (file)	jo-2009-11-24.E01	4474b818d2ad84beb74fbd7077d59e97
Primary disk (container)	(EnCase content)	274ada9b6b120f9b7a2192c27e115244
RAM	jo-2009-11-16.mddramimage	50c49185e254751caa0a9eadfa8fc635
Favourites USB (content)	(EnCase)	ad1d03cbdb8d81e918899479360f50dc
Work USB (content)	(EnCase)	8f23279deb398c3245829a98bc8fc1bd
Work USB carved	._TiggerTheCat copy.m4v (4KB)	edeca2340b4125f4ef4c949f9190ba5c
Work USB allocated	TiggerTheCat copy.m4v	9bd2dd0f202046eb032bdf46ba91fefa
Work USB carved	._SC00013.JPG	8d1eef424b0e5756ba56eb9d30bb579
Primary disk	Pics\patent05.JPG	ac0aceb421b7aa71b878ecc6bf946dad

The macOS resource-fork (._TiggerTheCat copy.m4v) MD5 matches between disk and Work USB → proves the USB transited Jo's macOS home machine. Main payload differs → the USB is a curated transfer device, not a full clone.

— End of Report —